lablab.ai x Alpaca AI Trading Agents Hackathon · 28 August to 4 September 2026

Delphi

A 0DTE iron condor agent on Alpaca paper trading

Short strikes are set by conformal risk control. The trade condition compares the certified cost of that interval with the market's price of the same interval.

Solo entry · MIT licence · github.com/jannissio/delphi-alpaca-agent

Competition paper account **PA31SEVJV9P9**, funded $100,000, options level 3, traded only by the submitted agent from 2026-09-03.

Reverse of a silver tridrachm of Delphi, c. 480–470 BC: four coffers, a dolphin in each. Two maxims stood at the temple, nothing in excess and know thyself. They correspond to the loss caps and to the claims this deck does not make.

Münzkabinett der Staatlichen Museen zu Berlin, inv. 18215461. Photograph Reinhard Saczewski. Public Domain Mark 1.0.

Premise: what the evidence supports

Over three trading sessions the profit and loss of an options strategy cannot be distinguished from noise. The agent is therefore built to produce evidence that needs no statistical power.

Traded index-option alphas have been indistinguishable from zero for fifteen years, and same-day premium selling is roughly zero after costs. The one public 0DTE rule study lost every positive net Sharpe ratio when its author corrected a hundredfold transaction-cost error in August 2026. [1][2][19]	**No edge is claimed, and the corrected result is cited against us.**
The half of the index-option premium that can be measured is earned overnight: delta-hedged S&P 500 returns to the buyer average about −1 % per day from close to open against +0.3 % from open to close. [3]	**Delphi never holds overnight, so it forgoes that half by construction. Pre-registered in advance, not explained afterwards.**
With three observations the Probabilistic Sharpe Ratio cannot reach 95 % at any performance level, and three perfect sessions at a credit of 0.20 of the wing cannot push an anytime-valid p-value below 0.51. [4][15]	**No Sharpe ratio, win rate or annualised return is reported anywhere in the repository.**

Reported instead: a bound on the expected payout of every trade, 31 logged risk gates, and a per-session ledger that `scripts/reproduce.py` regenerates and marks MATCH or MISMATCH. [R11]

[1] Dew-Becker & Giglio 2025. [2] Vilkov 2026 and its KNOWN-ISSUES, 2026-08. [3] Muravyev & Ni 2020 JFE. [4] Bailey & Lopez de Prado 2012. [15] Waudby-Smith & Ramdas 2023. [19] Almeida, Freire & Hizmeri 2025. In the repository: docs/THEORY.md sections 8 and 9, docs/evidence.md, README.md 'Why this design'. [R11] scripts/reproduce.py.

Setting the strikes: conformal risk control

The short strikes sit at the smallest interval whose expected payout to the buyer is certified, in finite samples, to be at most 10 % of the wing.

1 **Score**

For every past session, how far SPY actually moved divided by the move the options implied. The same unit in history and live.

2 **Loss**

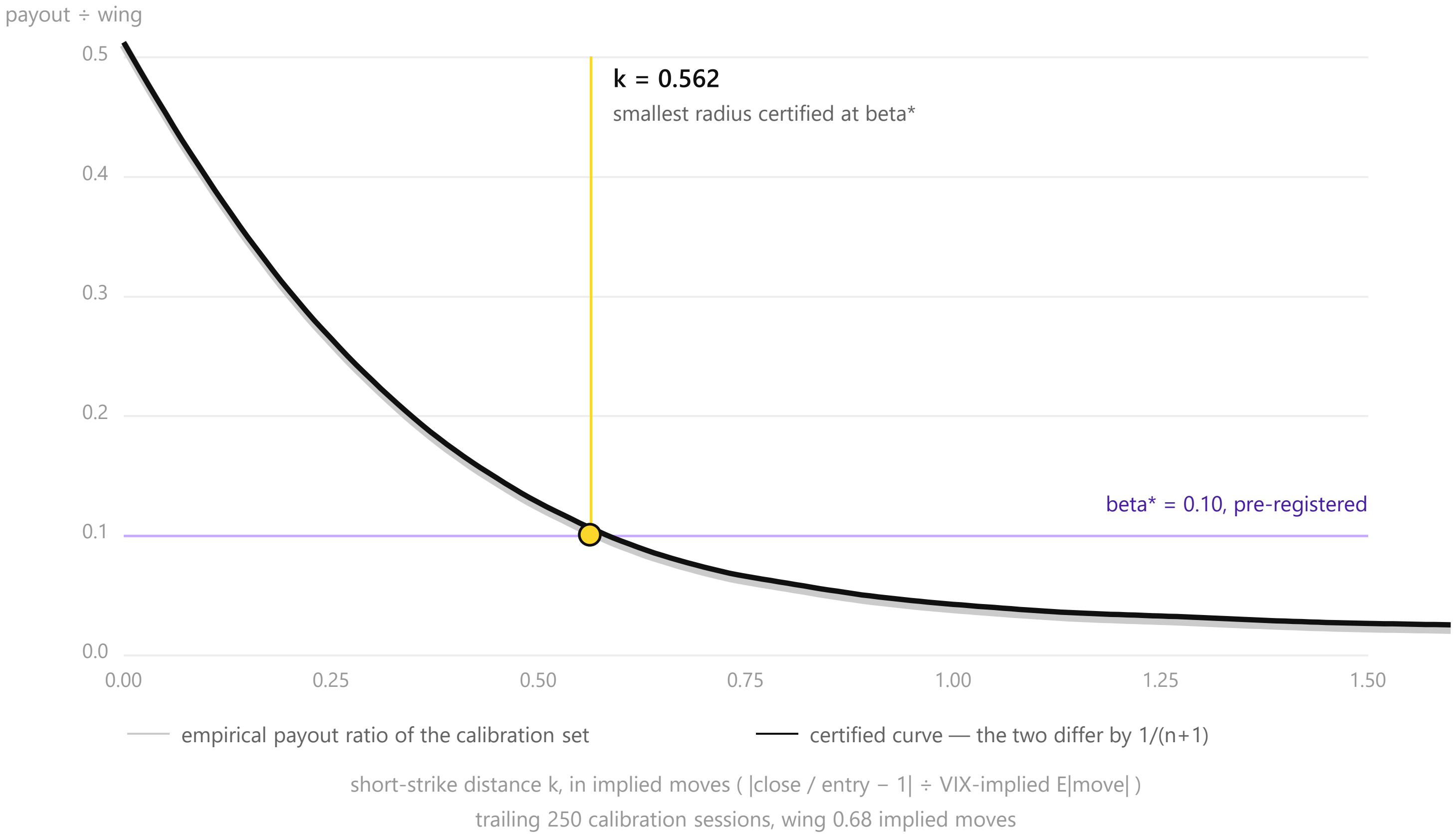
A condor pays its buyer min((r − k)+, w) / w, a loss between 0 and 1 that falls as the interval widens. [R1, Lemma 1]

3 **Radius**

From the trailing 250 scores, the smallest radius whose finite-sample corrected payout ratio is at most beta* = 0.10. [5][R1, Theorem 3]

4 **Level**

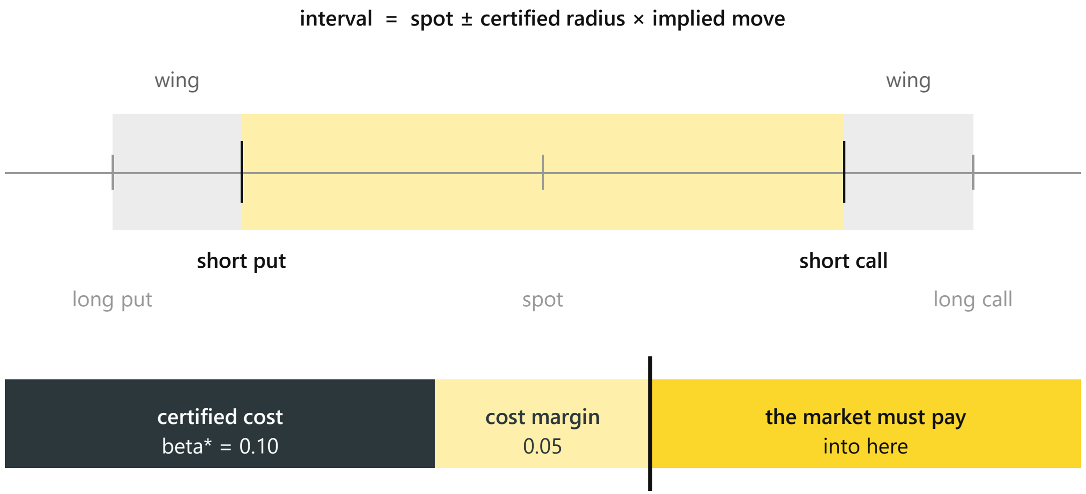
After every session, traded or not, an online level may tighten the radius and never widen it. [6][R1, Theorem 4]



[5] Angelopoulos, Bates, Fisch, Lei & Schuster, ICLR 2024. [6] Feldman, Ringel, Bates & Romano, TMLR 2023. In the repository: [R1] docs/THEORY.md Lemma 1, Theorems 3 and 4; docs/risk_curve.svg, regenerated by scripts/risk_curve.py; [R10] research/G_conformal_condor.md.

The trade condition: market price against certified cost

Sell only when the market pays at least the certified cost of the interval plus the modelled cost of trading it: credit / wing ≥ 0.10 + 0.05.



The horizontal axis is the credit as a fraction of the wing. The market's price of the interval, credit / wing, is read from the quote at the expected fill; the mid is logged alongside.

Back-fill, 618 sessions. History carries no option prices, so only the certified side can be replayed. [R2]

realised payout ratio vs 0.10	all	2024	2025	2026
conformal radius	**0.079**	0.070	0.079	0.090
fixed 1.10x rule	0.104	0.119	0.113	0.073

Why credit / wing is the right comparison

For a vertical spread the credit divided by the width is the risk-neutral survival probability averaged across the wing band. It is the market's price of exactly the event the certificate bounds. [7][R1, Lemma 2]

What the condition buys

One line, no free parameter. Every trade that passes is certified, in expectation and under the stated assumptions, not to lose after the modelled round-trip cost of four legs. A trade on the boundary expects its cost and no more.

What it does not buy

The 0.05 is a cost margin, not a profit target, and the certificate is marginal while the condition itself selects sessions. If the market never pays, the agent never trades and the ledger records a closed gate.

[7] Breeden & Litzenberger 1978. In the repository: [R1] docs/THEORY.md Lemma 2 and gate 31; [R2] docs/conformal_backfill.md, 618 calibrated sessions 2024-01-02 to 2026-09-01; [R5] docs/CONFIG_CHANGES.md, credit read at the expected fill from 2026-09-02.

Categories from the model, numbers from code

The language model returns enumerations and may only reduce risk. Deterministic code sets every number, and a test proves the order is byte-identical across different model outputs.

Inputs	DATA	Alpaca option chain snapshots, IEX bars and news; Cboe VIX, VIX3M and VIX1D.
Regime	LLM	Enums only: VOL_REGIME, TREND, EVENT_RISK, strategy family, veto. Tickers and dates masked [8]. Three calls must agree, which raises the NO_TRADE rate rather than accuracy [9].
Strategy	CODE	Certified radius to short strikes and wings; credit / wing read from the quote.
Sizing	CODE	2 % of capital per session, 6 % across the campaign; a logistic regime model trained on 9,230 sessions may only shrink the size, never raise it.
Gates	CODE	31 checks, evaluated on every candidate order and logged pass by pass; gate 31 is the trade condition.
Critic	LLM	Sees the finished, gate-validated order and may say PASS, REDUCE or BLOCK. It can never enlarge.
Execution	CODE	Multi-leg limit ladder priced in ticks and re-quoted at every rung; flat by 15:15 ET.
Record	CODE	Append-only audit log with git and configuration hashes, from which the report, the public ledger and the determinism and leakage audits are computed.

Sampling is pinned on temperature, top_p, top_k and seed, and per-field decision entropy is logged in bits [10]. Determinism is an assurance property, not a correctness one: 1,000 temperature-zero completions of one prompt have been measured to give 80 distinct outputs [11]. Open-weight models through Featherless.ai — DeepSeek-V3.2 for regime and critic, Qwen3-30B-A3B for the journal.

[8] Glasserman & Lin 2024. [9] Kim 2026. [10] Koviazin, Mudarisov, Polyachenko & State, IC-AIF 2026. [11] Thinking Machines Lab 2025. In the repository: README.md 'Architecture', docs/WRITEUP.md 'AI logic', [R7] docs/report_2026-09-02.md (LLM section).

Risk gates and execution

Every candidate order passes 31 gates before it is sent, and orders are priced in ticks and re-quoted at every rung.

Thirty-one gates [R4][12]

- 30 controls derived from SEC Rule 15c3-5, MiFID II RTS 6, FINRA Notice 15-09 and the SEC's Knight Capital order, plus gate 31, the trade condition.
- Session loss capped at 2 % of capital with a continuous taper toward a 6 % campaign cap; $1,000 of maximum loss and 5 contracts per order; defined risk only, so every short leg has a bought wing in the same package.
- Entry windows by weekday; Friday is a logged NO_TRADE day because of the payrolls release at 08:30 ET; no entry after 14:00; flat by 15:15 ET on liquidity grounds.
- Kill switch checked every cycle with cancel-all under five seconds, a daily loss kill on an independent code path, and order and position reconciliation every cycle. A mismatch halts new risk, and on the pilot day one did — on a book that was correct. alpaca-py returns the side as `PositionSide.LONG`, the comparison was case-sensitive, so every bought wing read as short. The control did its job; the defect was ours, and it is fixed. [R7][R13]

Execution [13][R5]

- Same-day quoted spreads are one or two ticks wide, so collars and ladders are expressed in ticks and never as a percentage of the mid. No market orders, ever.
- Alpaca paper fills only marketable limit orders. On the pilot day three rungs fixed at 0.49, 0.48 and 0.47 expired unfilled while the package's natural price fell from 0.47 to 0.39 in sixty seconds.
- Since then every rung is priced from quotes pulled at send time and reaches the live natural within sixty seconds, and no rung may go below the credit floor the gates approved.
- Every parameter changed after the first live cycle is listed with its date, its evidence and its effect. [R5]

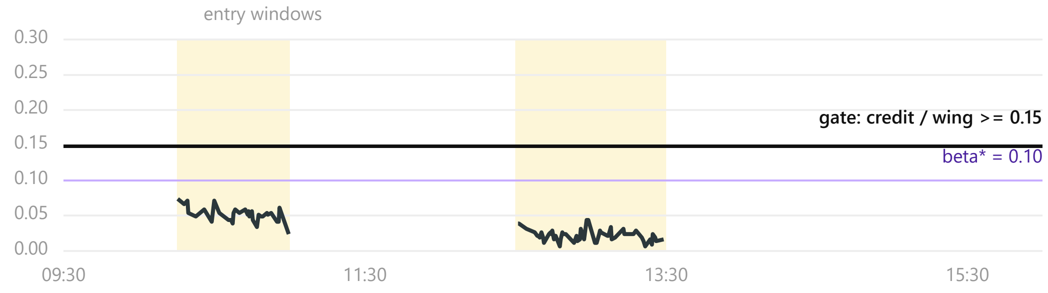
On Alpaca	Trading API (alpaca-py)	Alpaca CLI	Alpaca MCP server
paper enforced at start-up	chains, multi-leg orders, positions, IEX bars for the calibration scores	independent monitoring and reconciliation path, JSON to disk	operator inspection of account and chains

[12] SEC Rule 15c3-5; MiFID II RTS 6; FINRA Notice 15-09; SEC order in re Knight Capital, 2013. [13] Fu, Li & Musto, SEC DERA. In the repository: [R4] config/risk_limits.yaml and agent/gates/engine.py; [R5] docs/CONFIG_CHANGES.md; [R7] docs/report_2026-09-02.md; [R13] agent/execution/recon.py and commit cb441a8.

Live operation and results

On the judged account the gate stayed closed all day. That is the mechanism. The pilot the day before, on a separate account, shows the path when it opens.

Competition account PA31SEVJV9P9 · 2026-09-03



What the market offered for the band, against the gate. 87 priced snapshots inside the two windows; the highest was 0.0725 of the wing.

Interval committed **10:15 ET**: spot 768.49, certified radius **$3.27**, wing $4. Equity unchanged at $100,000.

315
chain evaluations, each one NO_TRADE

0
trades; no order was ever sent

0.005 – 0.0725
offered, of the wing, against a gate of 0.15

0.338
of the wing, paid out at the close

SPY closed **outside** the certified interval. Sold at the best price the market offered all day, the band would have cost about **$106** a contract. On this day, the refusal was the right decision.

Development pilot PA314NYH4H7G · 2026-09-02 · not counted

- **Regime** — three votes agreed, 3 : 0, on IRON_CONDOR_0DTE, one contract
- **Gates** — 20 of 20 passed
- **Critic** — PASS
- **Ladder** — 0.45 → 0.44, filled at 0.44 — 0.147 of the wing
- **Halt** — reconciliation mismatch: a false positive, found and fixed
- **Exit** — take profit at 0.30 · +$14

The dashboard and a full cycle of the audit log are shown in the video, not on this slide.

Under the rule now running, this day would not have traded either: **0.147 < 0.15**.

In the repository: [R14] docs/report_2026-09-03.md — the judged session: chain 315, no_trade 315, interval committed 14:15:25 UTC; after the close, realised 0.788 vs certified 0.557, payout 0.338, beta_t 0.1647 to 0.1635, alpha_t 0.219 to 0.215, risk e-process unchanged. [R15] docs/band_price_2026-09-03.json — the price per snapshot. [R7] docs/report_2026-09-02.md — the pilot. [R8] docs/ledger.json.

Scope of the claim, and next steps

What is machine-checked, what is assumed, what is not claimed, and what more data would change.

Proved

13 theorems in Lean 4 with Mathlib, no `sorry`, on the three standard axioms.

The condor loss lies in [0, 1] and is non-increasing in the radius; the payoff is affine in the loss; and if the average calibration loss is at most beta and the credit is at least (beta + margin) x wing, the average payoff is at least margin x wing.

Option pricing has been formalised before and we cite it. A machine-checked risk-control lemma for a trading decision rule we did not find. [R6][18]

Assumed

Exchangeability of the calibration scores.

The certificate is marginal, while the trade condition selects sessions through a quantity that depends on the certified radius. Validity conditional on the gate opening does not follow; the exact repair needs a credit history the basic data plan does not have. [17]

The e-process null is conditional and therefore stronger than the theorem. [R1, section 6 and Theorem 3, remark v]

Not claimed

No Sharpe ratio, win rate or annualised return. The Sharpe-maximising manipulation is precisely our instrument, so the number would be uninformative even if it were computable. [4][14]

No intraday variance risk premium. The measurable half is overnight and is forgone by construction. [3][R1, section 8]

Not the first conformal trading decision: that prior art exists and is cited. [16]

The online level currently has no effect on the interval and is a one-sided safety valve.

Next

A 1DTE overnight condor entered near 15:00 ET, calibrated on the close-to-close horizon where the regime model has fifteen validation blocks.

An option-price history, so the market side of the ledger can be back-tested rather than only logged.

The 10:30 series extended to 2018, so the model comparison gains power.

Mondrian conditioning on the gate event; five votes instead of three.

[4] Bailey & Lopez de Prado 2012. [14] Goetzmann, Ingersoll, Spiegel & Welch 2007. [16] Lekeufack et al., ICRA 2024; Ryan 2026. [17] Jin & Ren 2025; Gibbs, Cherian & Candes 2025; Xu, Guo & Wei 2025. [18] Ushakov & Berdinsky 2026; Coelho 2026. In the repository: [R6] lean/Delphi/Condor.lean and lean/README.md; [R1] docs/THEORY.md sections 6 and 8; [R12] research/STATE_OF_THE_ART.md section 14.

Backup: the certified interval replayed through history

618 calibrated sessions, 2024-01-02 to 2026-09-01. The realised payout ratio stays below the certificate in every year; the fixed-strike rule drifts with whatever the year's realised-to-implied ratio happens to be.

	sessions	realised payout ratio	target	mean radius	fixed rule 1.10x	coverage
All calibrated sessions	618	**0.0792**	0.10	0.809	0.1039	0.806
2024	202	0.0703	0.10	0.972	0.1188	0.832
2025	249	0.0791	0.10	0.795	0.1126	0.799
2026	167	0.0901	0.10	0.630	0.0729	0.784
Since 2024-12-30 (research/G sample)	418	0.0831	0.10	0.730	0.0962	0.792

State after the back-fill

Online risk level beta = 0.1642 and coverage level alpha = 0.2180 after 618 sessions, window 250 scores, clip [0.35, 1.60], step 0.005.

Because the realised payout ratio has run below beta*, the online level has drifted above it: today the online layer is a one-sided safety valve with no effect on the interval, which we report rather than dress up.

Two limits of this evidence

The whole calibration window lies inside the 2024 to 2026 regime, so it is not a test across regimes.

The coverage track — controlling how often the close leaves the interval — is kept only as the logged counterfactual: most breaches are partial, and the break-even inside probability is 75.7 %, not 83 %.

Repo: [R2] docs/conformal_backfill.md, replayed 2026-09-02; [R10] research/G_conformal_condor.md. Score = |close / price at 10:30 − 1| divided by the VIX-implied expected absolute daily move; wing 0.50 % of spot. No option prices exist in this history, so the market side of the ledger cannot be back-tested.

Backup: the regime model and its negative results

A twelve-coefficient logistic regression that can only shrink the position. Two 2026 tabular foundation models ran the same protocol and did not beat it.

What it is

9,230 sessions assembled from Cboe (S&P 500 since 1975, VIX since 1990) and Alpaca IEX (SPY open and close from 2020-07, 30-minute bars from 2024-01). A standardised logistic regression on 1,533 SPY sessions maps regime features known at the morning entry to the probability that the session closes inside the short strikes.

What it may do

Set the size multiplier to 1, 0.5 or 0 — full at or above the historical tercile threshold, half below it, zero in the bottom decile. It can never increase a position.

What it cannot do

On the exact 10:30-to-close horizon the logit is worse than a constant, with two year-blocks and no inference possible. On the close-to-close horizon, with fifteen blocks, it is the best single model. Both results are reported.

What lost

XGBoost, LightGBM, a logit and XGBoost average, TabPFN v2 and TabICL v2, run under the identical expanding-window protocol: seven families across three horizons, 21 configurations. The foundation models win only where nothing can be measured and lose, with bootstrap intervals excluding zero, where it can.

What the back-test cannot settle

It assumes a constant credit of 17 % of the wing, which invalidates any conclusion that changes the interval width. The live agent reads the credit off the quote instead.

The null it supplies

The same history gives the random-entry Monte Carlo null against which campaign profit and loss is reported as a percentile rather than as alpha: median +259 dollars, 5th percentile −503 dollars over 2.5 sessions at two contracts per session.

Repo: [R9] docs/regime_model_report.md and research/H_tabular_ml_small_data.md, 21 configurations; README.md 'What was trained on history'. Data: scripts/history_data.py.

Backup: what three sessions can and cannot show

Two test supermartingales are computed from the ledger and reported, never used to halt. The ceiling on the second one is arithmetic, and it is the reason no Sharpe ratio appears anywhere.

Risk process — evidence against the certificate

Null: the expected payout ratio given the past is at most beta* = 0.10. A p-value below 0.05 would reject the certificate. It does not.

	sessions	mean payout ratio	anytime p
All	619	**0.079**	**0.376**
2024	202	0.070	0.376
2025	249	0.079	0.143
2026	168	0.090	1.000

Running maximum of the wealth process 2.659 over 619 sessions. A conformal test martingale on clean data has been observed to fire in 135 of 135 runs at the 5 % level, so this process is reported and never used to halt.

Profit process — and its ceiling

T maximal wins at a credit of g times the wing cannot produce more than (1/(1−g))^T. The smallest attainable anytime-valid p-value is its reciprocal.

credit/wing	T = 1	T = 3	T = 5	T = 10	sessions for p <= 0.05
0.15	1.18	1.63	2.25	5.08	19
0.20	1.25	**1.95**	3.05	9.31	**14**
0.25	1.33	2.37	4.21	17.76	11

At a credit of 0.20 of the wing, three perfect sessions reach 1.95, so the p-value cannot fall below 0.51. Fourteen consecutive perfect packages would be needed for 0.05. Evidence accrues per package, so two smaller packages a session raise the three-session ceiling to 3.81 at an unchanged risk budget.

Repo: [R3] docs/evidence.md, regenerated by scripts/evidence.py; [R1] docs/THEORY.md section 9. [15] Waudby-Smith & Ramdas 2023; Ramdas, Grunwald, Vovk & Shafer 2023. Bets pre-registered at lambda = 1.0 and eta = 1.0.

Sources: literature (1 of 2)

Not part of the presentation itself. The numbers in square brackets on the slides refer to these entries, so that every claim can be traced back to where it comes from.

[1] Dew-Becker, I. & Giglio, S. (2025). The Decline of the Variance Risk Premium: Evidence from Traded and Synthetic Options. Federal Reserve Bank of Chicago WP 2025-17.

[2] Vilkov, G. (2026). 0DTE Trading Rules (SPXW, n = 1,319), with the author's KNOWN-ISSUES record of the August 2026 transaction-cost correction, github.com/vilkovgr/0dte-strategies.

[3] Muravyev, D. & Ni, X. (2020). Why do option returns change sign from day to night? Journal of Financial Economics 136(1):219-238.

[4] Bailey, D. & Lopez de Prado, M. (2012). The Sharpe Ratio Efficient Frontier. Journal of Risk 15(2) - the Probabilistic Sharpe Ratio and the minimum track record length, which is the claim made here. Their Deflated Sharpe Ratio (2014, JPM 40(5)) addresses a different problem, selection across many trials.

[5] Angelopoulos, A. N., Bates, S., Fisch, A., Lei, L. & Schuster, T. (2024). Conformal Risk Control. ICLR 2024; arXiv:2208.02814.

[6] Feldman, S., Ringel, L., Bates, S. & Romano, Y. (2023). Achieving Risk Control in Online Learning Settings. TMLR; arXiv:2205.09095.

[7] Breeden, D. T. & Litzenberger, R. H. (1978). Prices of State-Contingent Claims Implicit in Option Prices. Journal of Business 51(4).

[8] Glasserman, P. & Lin, C. (2024). Assessing Look-Ahead Bias in Stock Return Predictions Generated by GPT Sentiment Analysis. Journal of Financial Data Science 6(1).

[9] Kim (2026). Are Diversity Metrics Measuring Diversity? arXiv:2607.20768 - majority voting beats the strongest member in 9.98 % of size-3 subsets. Bahuguna (2026). arXiv:2608.11403 - majority voting lowers accuracy on hard tasks. Zhang et al. (2026). arXiv:2608.18795 - agreement is graded evidence, not certification.

[10] Koviazin, Mudarisov, Polyachenko & State (2026). IC-AIF 2026, DOI 10.1145/3800973.3801029.

Sources: literature (2 of 2)

Regulation is taken as a design template for the gates; no compliance is claimed. Entries [16] to [18] are prior art that this work does not claim as its own.

[11] Thinking Machines Lab (2025), on inference kernels that are not batch-invariant.

[12] SEC (2010). Rule 15c3-5, Release 34-63241. ESMA / EU (2017). MiFID II RTS 6, Delegated Regulation 2017/589, Articles 12, 15 and 17. FINRA (2015). Regulatory Notice 15-09. SEC (2013). In the Matter of Knight Capital Americas LLC, Release 34-70694. Taken as a design template; no compliance is claimed.

[13] Fu, L., Li, S., Musto, D. K. & Pearson, N. D. (2025). Hope at a Reasonable Price: Customer Use of Limit Orders in the 0DTE Market. SEC DERA Working Paper, 16 March 2025.

[14] Goetzmann, W., Ingersoll, J., Spiegel, M. & Welch, I. (2007). Portfolio Performance Manipulation and Manipulation-proof Performance Measures. RFS 20(5):1503-1546.

[15] Waudby-Smith, I. & Ramdas, A. (2023). Estimating means of bounded random variables by betting. JRSS-B 86(1). Ramdas, A., Grunwald, P., Vovk, V. & Shafer, G. (2023). Game-theoretic statistics and safe anytime-valid inference. Statistical Science 38(4).

[16] Lekeufack, J., Angelopoulos, A. N., Bajcsy, A., Jordan, M. I. & Malik, J. (2024). Conformal Decision Theory. ICRA 2024. Ryan (2026). Conformal Kelly. arXiv:2608.01494. Prior art we do not claim.

[17] Jin, Y. & Ren, Z. (2025). JRSS-B 87(4). Gibbs, I., Cherian, J. J. & Candes, E. J. (2025). JRSS-B 87(4). Xu, Guo & Wei (2025). Selective Conformal Risk Control. arXiv:2512.12844.

[18] Ushakov & Berdinsky (2026). arXiv:2608.19223. Coelho (2026). arXiv:2606.01356. Echenim, Guiol & Peltier (2018). arXiv:1807.09873. Prior formal verification in finance.

[19] Almeida, C., Freire, G. & Hizmeri, R. (2025). 0DTE Asset Pricing. SSRN 4701401. Beckmeyer, H., Branger, N. & Gayda, L. (2023). Retail Traders Love 0DTE Options... But Should They? SSRN 4404704.

Sources: this repository

Every figure on the slides is regenerated by a script in the repository; scripts/reproduce.py recomputes all of them and prints MATCH or MISMATCH against the documents.

[R1] docs/THEORY.md — definitions, Lemmas 1, 2 and 5, Theorems 3 and 4, the assumption list in section 6, the pre-registered give-up of the overnight premium in section 8, the e-processes in section 9.

[R2] docs/conformal_backfill.md — 618 calibrated sessions replayed through both online tracks.

[R3] docs/evidence.md — the two e-processes and the evidence ceiling.

[R4] config/risk_limits.yaml and agent/gates/engine.py — the 31 gates as code, immutable at runtime.

[R5] docs/CONFIG_CHANGES.md — every pre-registered parameter changed after the first live cycle, with date, evidence and effect.

[R6] lean/Delphi/Condor.lean and lean/README.md — 13 theorems, no sorry, and the narrow statement of what is new.

[R7] docs/report_2026-09-02.md — the pilot session, on the development account PA314NYH4H7G.

[R8] docs/ledger.json and docs/index.html — the published per-session ledger.

[R9] docs/regime_model_report.md and research/H_tabular_ml_small_data.md — 21 model configurations and their negative results.

[R10] research/G_conformal_condor.md — the study behind the conformal condor, including the refutation of the width-changing variant.

[R11] scripts/reproduce.py — regenerates every number quoted in the documents and prints MATCH or MISMATCH.

[R14] docs/report_2026-09-03.md and docs/report_2026-09-03.json - the judged session: 315 chain evaluations, 315 NO_TRADE, no order sent, equity unchanged; and, after the close, the realised move against the certified radius with the two levels it moved.

[R15] docs/band_price_2026-09-03.json - what the market offered for the band at each snapshot, and docs/band_price_2026-09-02_pilot.json for the pilot day.

[R13] agent/execution/recon.py - the reconciliation gate, with the pilot-day false positive and its cause in the comment; fixed in commit cb441a8.

[R12] research/STATE_OF_THE_ART.md — the synthesis, and section 14, the independent literature audit of 2026-09-02.